

Ex.no:8	MINI PROJECT – SOCIAL MEDIA ANALYTICS

Domain and Problem Statement

Domain: Social Media Analytics & Instagram Performance Analysis

Problem Statement:

Businesses and content creators generate large volumes of Instagram data but often lack clear insights into content performance, audience engagement, reach, and follower growth. This project addresses these challenges by transforming raw Instagram post data into structured business intelligence using Power BI and Tableau, enabling analysis of engagement trends, content performance, posting patterns, traffic sources, and overall Instagram performance to support data-driven content and marketing decisions.

Dataset Description

Dataset Name: Instagram Analytics Dataset

Granularity: Each row represents a single Instagram post.

Timeframe Covered: 2025

Key Entities & Attributes:

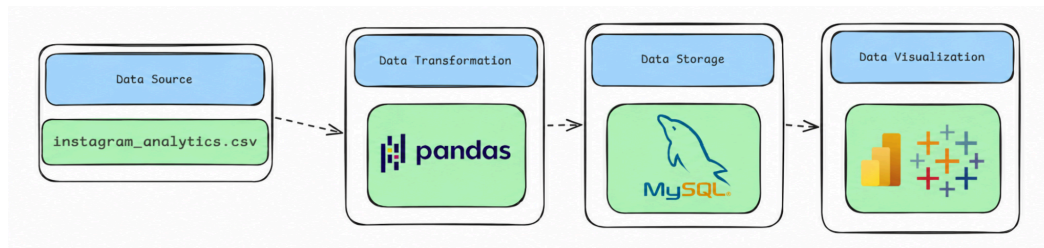
- **Account & Content:** account_id, account_type, media_type, content_category, caption_length, hashtags_count, has_call_to_action.
- **Post Details:** post_id, post_date, post_datetime, post_hour, day_of_week.
- **Engagement Metrics:** likes, comments, shares, saves, engagement_rate.
- **Reach & Growth:** reach, impressions, followers_gained, traffic_source.
- **Performance:** performance_bucket_label.

Data Source:

Primary Source: Instagram Analytics Dataset prepared for social media performance analysis.

Storage Infrastructure: Cleaned and pre-processed Instagram post records maintained in an Excel dataset and imported into Microsoft Power BI and Tableau for interactive dashboard development and analysis.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

df = pd.read_csv("instagram_analytics.csv")

# Create a copy for processing
instagram = df.copy()

# Handle missing values
instagram["account_type"] = instagram["account_type"].fillna("Unknown")
instagram["media_type"] = instagram["media_type"].fillna("Unknown")
instagram["content_category"] = instagram["content_category"].fillna("Unknown")
instagram["traffic_source"] = instagram["traffic_source"].fillna("Unknown")

# Replace missing numerical values with 0
numeric_columns = [
    "likes",
    "comments",
    "shares",
    "saves",
    "reach",
    "impressions",
    "followers_gained"
]

for column in numeric_columns:
    instagram[column] = instagram[column].fillna(0)

# Remove duplicate records
instagram.drop_duplicates(inplace=True)

# Convert date column
```

```

instagram["post_date"] = pd.to_datetime(
    instagram["post_date"], errors="coerce"
)
instagram.columns = (
    instagram.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)
# Display dataset information
print("Original Shape:", df.shape)
print("Cleaned Shape:", instagram.shape)

print("\nDataset Preview:")
print(instagram.head())

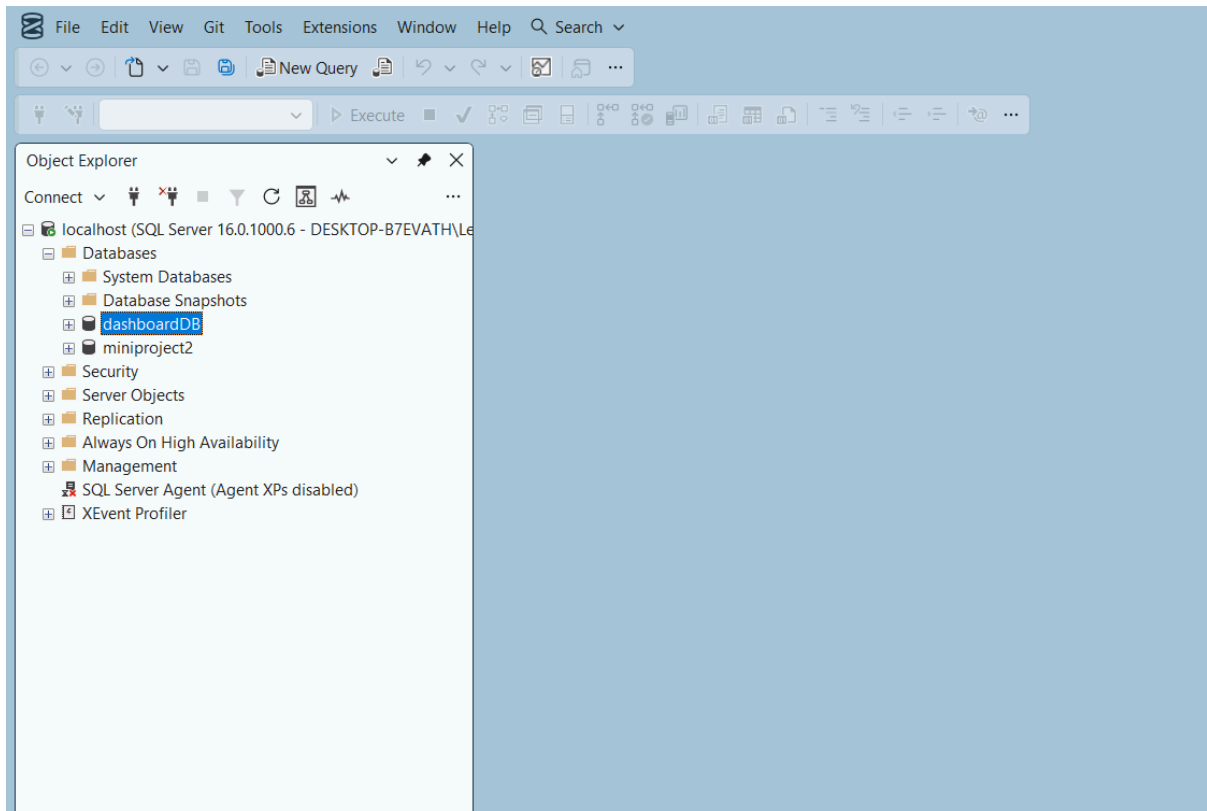
print("\nDataset Information:")
print(instagram.info())

```

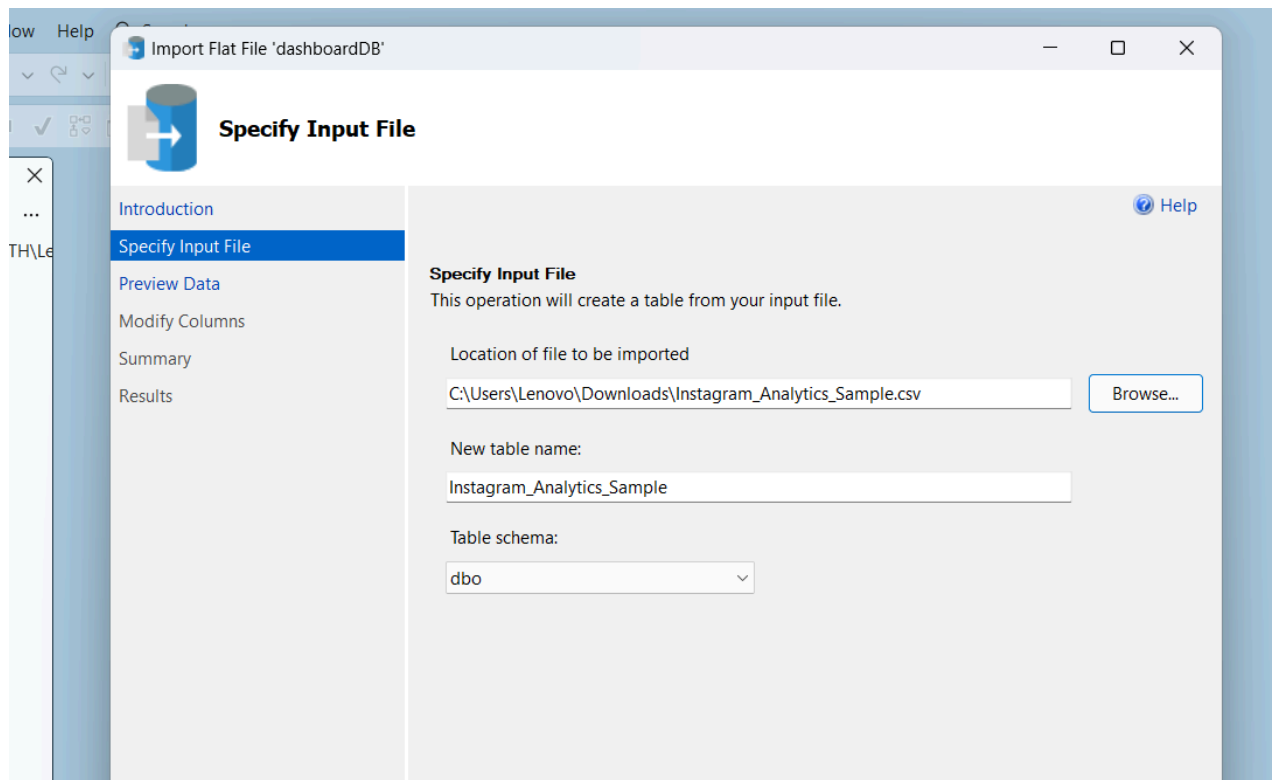
2. SQL SERVER CONNECTIVITY

INSTALL SQL SERVER MANAGEMENT STUDIO

1) Create database named “dashboardDB”

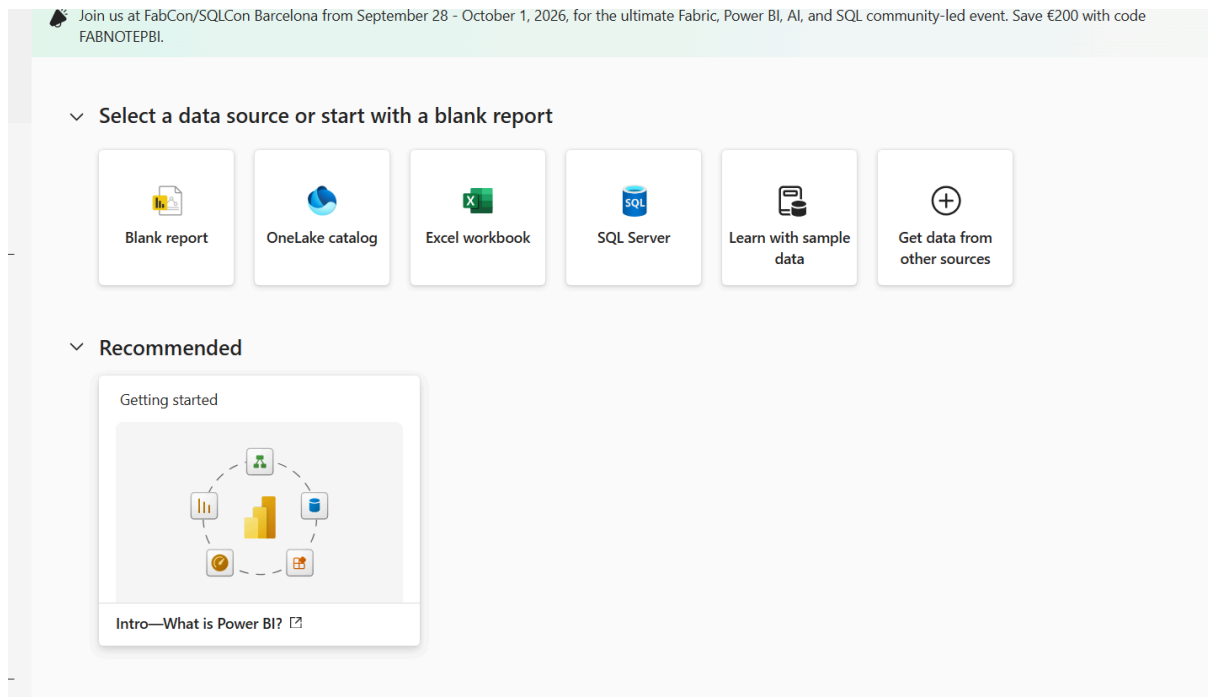


2) Import the csv file of the dataset

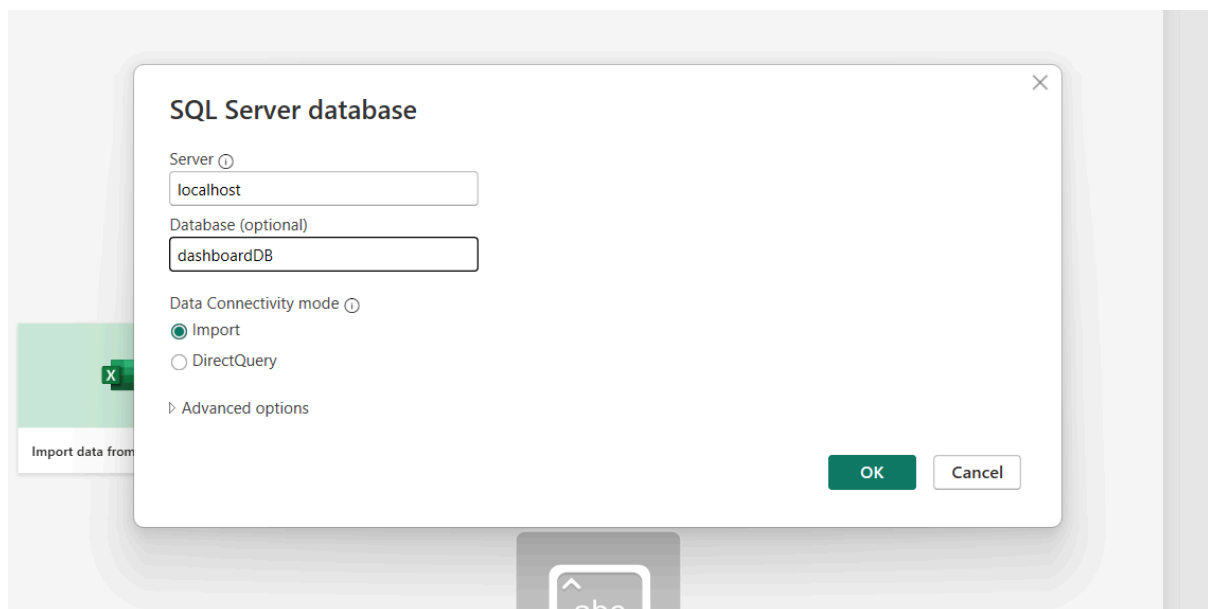


POWER BI

3) Now go to MICROSOFT POWER BI



4) Click SQL SERVER



5) Click OK and click transform data

TXN000000023	CUST018432	01-01-2019 00:00:00	21:28	Cu
--------------	------------	---------------------	-------	----

< >

Load Transform Data Cancel

6) Create DAX measures

TOTAL LIKES :

```
1 Total Likes = SUM('Instagram_Analytics'[likes])
```

TOTAL POSTS:

```
1 Total Posts = COUNT('Instagram_Analytics'[post_id])
```

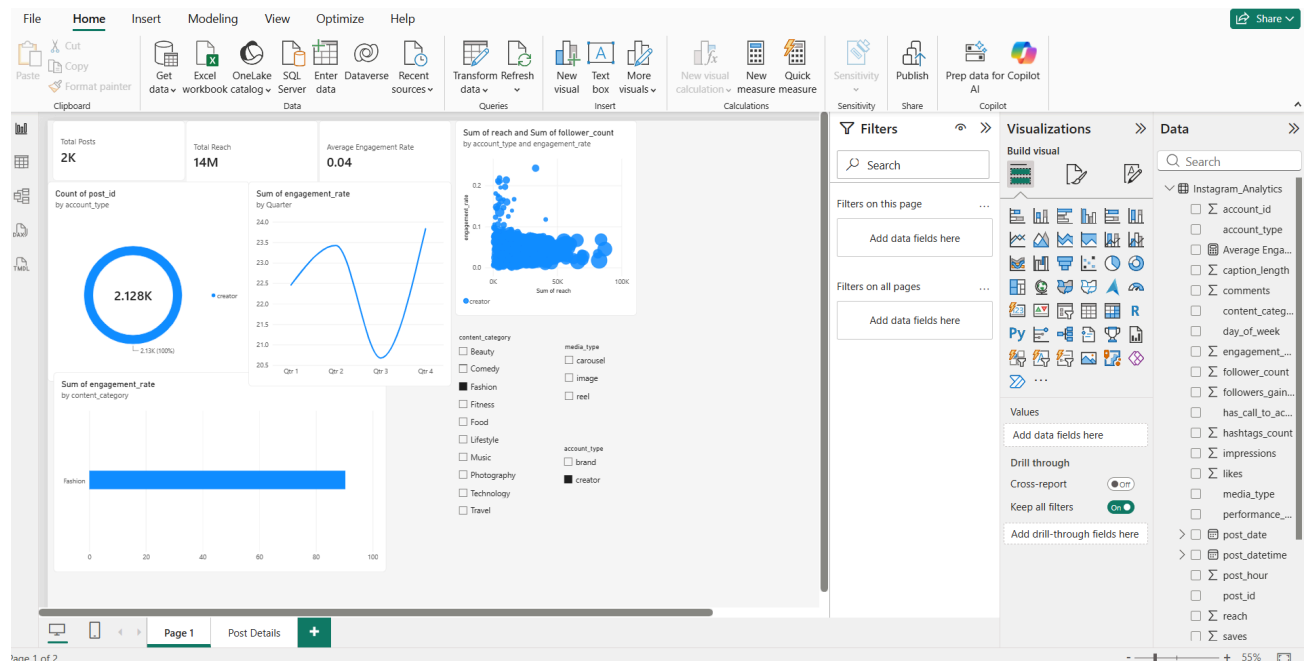
TOTAL REACH

```
1 Total Reach = SUM('Instagram_Analytics'[reach])
```

AVERAGE ENGAGEMENT RATE

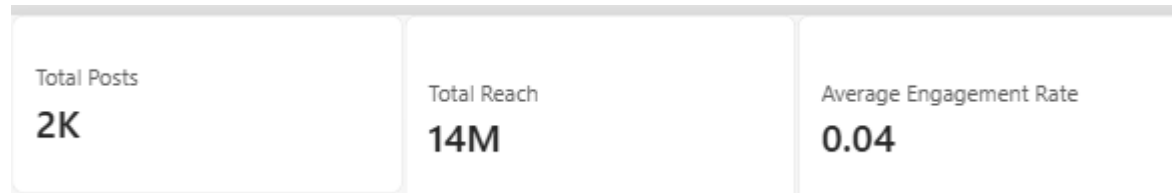
```
1 Average Engagement Rate = AVERAGE('Instagram_Analytics'[engagement_rate])
```

7) Power BI Final Executive Dashboard & Drill-Through

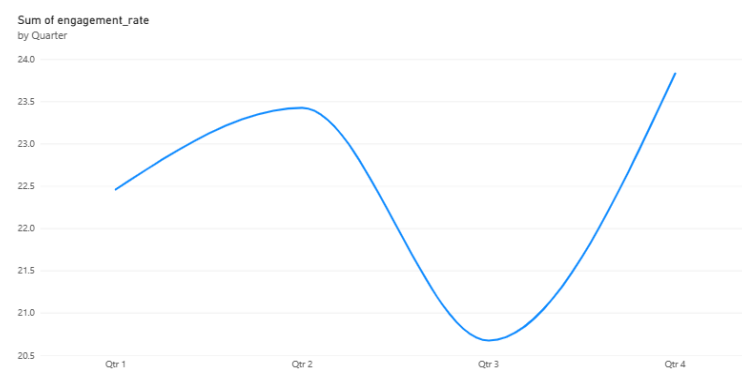


1. Executive Canvas:

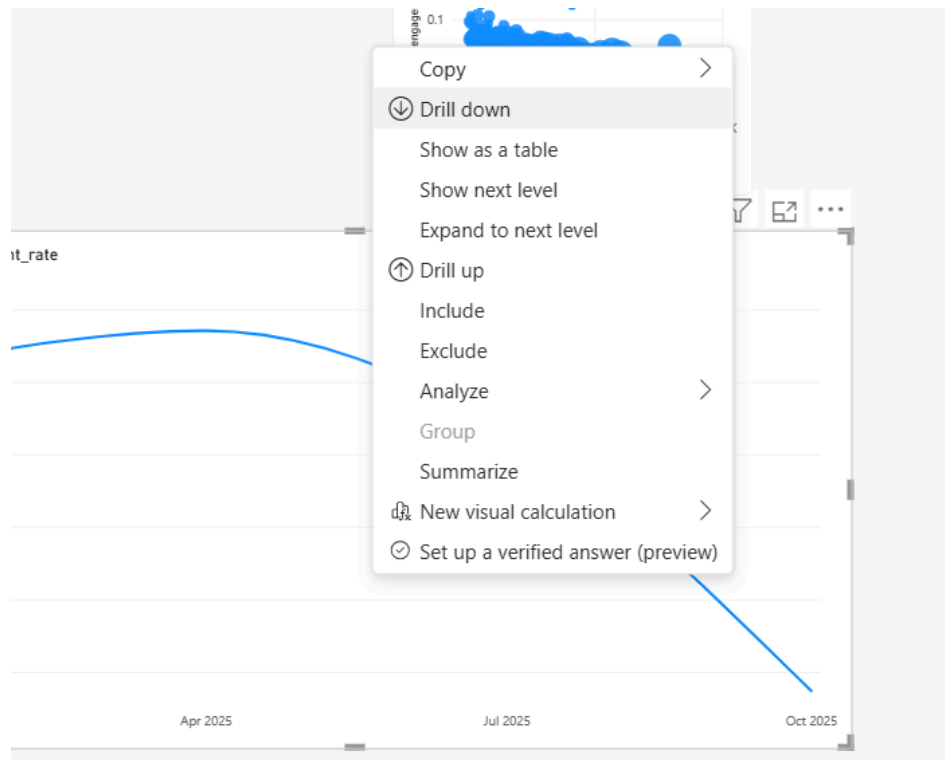
Place Card Visuals across the top: Total Likes, Total Transaction Reach, Total Shares, Average Engagement Rate



Add a Line Chart for sum of engagement rate .

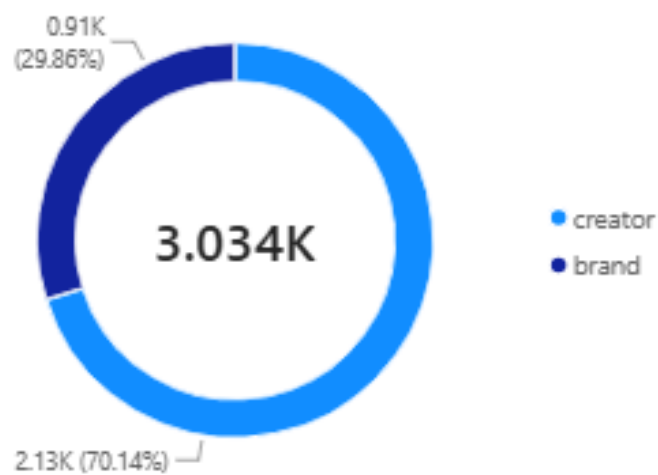


Enabled Drill Down

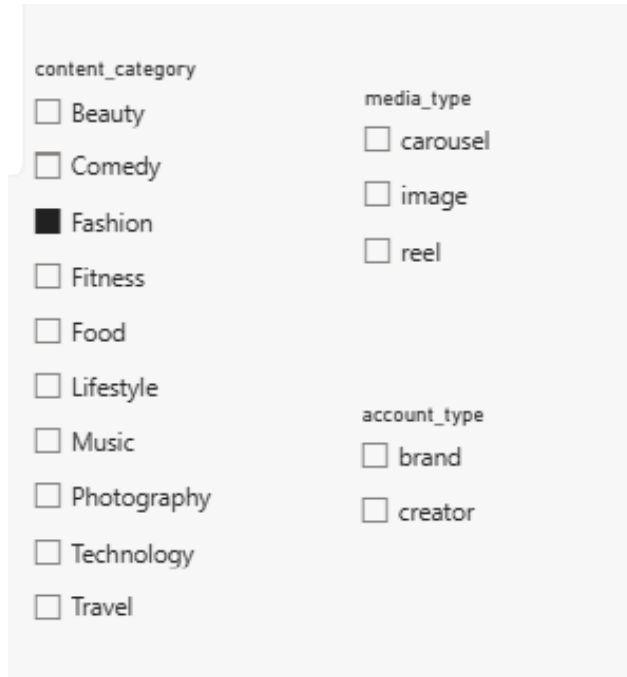


count by post id visualised by donut chart

Count of post_id
by account_type

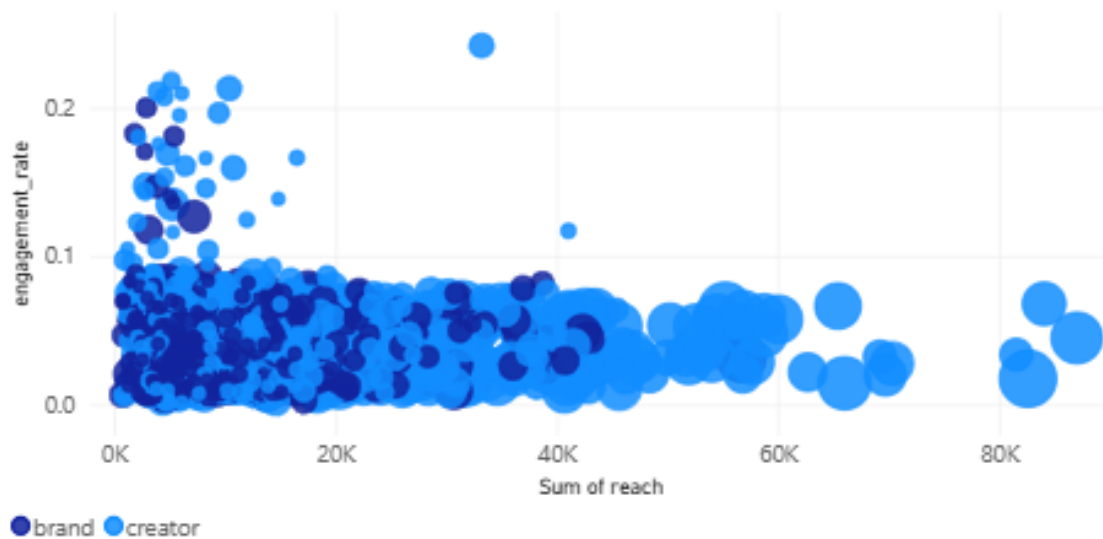


Create 3 slicers based on account type, content category and media type



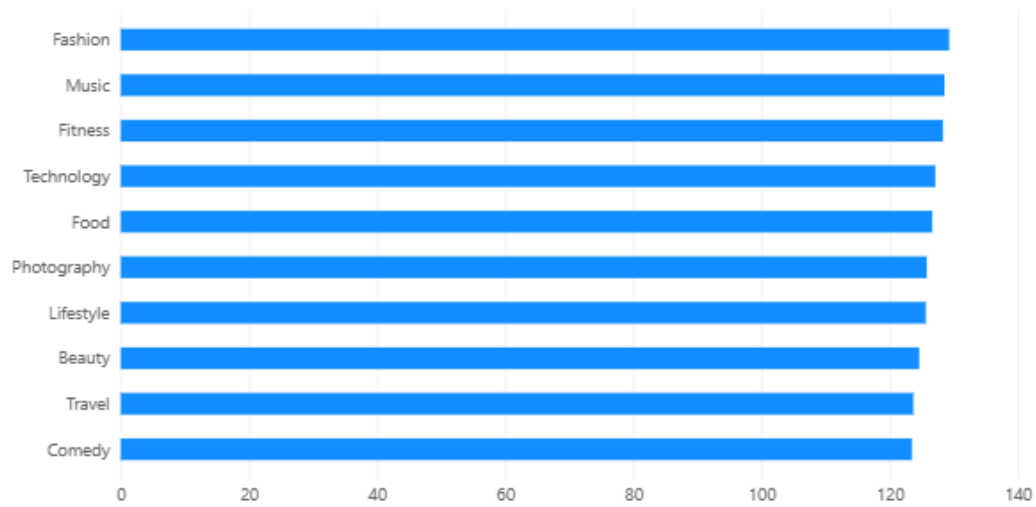
Scatter plot for sum of reach and sum of follower count

Sum of reach and Sum of follower_count
by account_type and engagement_rate



sum of engagement rate by content category

Sum of engagement_rate
by content_category



Enable drill-through

Values

Add data fields here

Drill through

Cross-report ☒

Keep all filters ☒

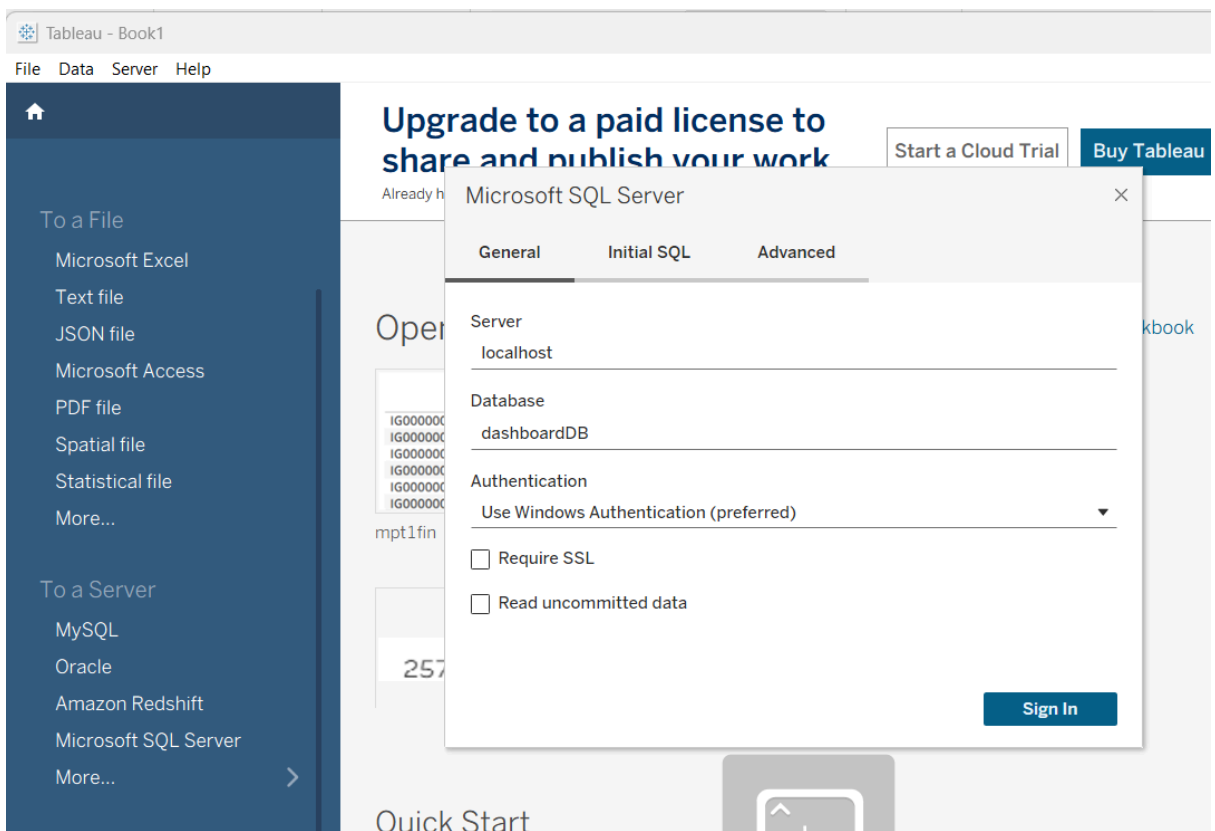
account_type ☐ ☐ ☐

is (All)

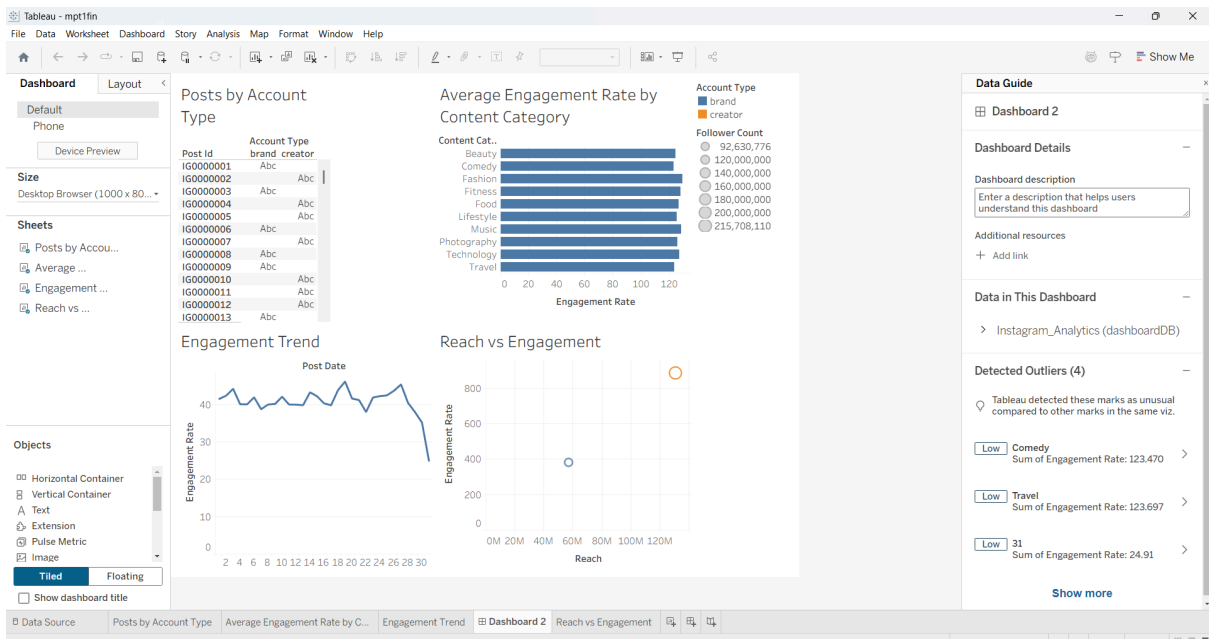
post_id	account_type	media_type	content_category	Sum of likes	Sum of comments	Sum of shares	Sum of reach	Sum of impressions	Sum of engagement_rate
IG0000001	brand	reel	Technology	194	5	7	4327	6230	0.04
IG0000002	creator	image	Fitness	449	10	21	7451	8268	0.07
IG0000003	brand	reel	Beauty	114	2	1	1639	2616	0.05
IG0000004	creator	carousel	Music	91	0	7	2877	3171	0.03
IG0000005	creator	reel	Technology	154	8	5	5350	8503	0.02
IG0000006	brand	carousel	Photography	149	3	4	2668	3472	0.05
IG0000007	creator	carousel	Food	371	9	14	5018	7671	0.06
IG0000008	brand	reel	Technology	18	1	0	1078	1357	0.02
IG0000009	brand	reel	Beauty	132	4	7	1900	2436	0.07
IG0000010	creator	reel	Music	101	2	3	2155	2942	0.04
IG0000011	creator	reel	Fitness	418	11	21	14307	20978	0.02
IG0000012	creator	carousel	Beauty	49	6	2	762	1098	0.05
IG0000013	brand	image	Beauty	111	1	7	4364	6329	0.02
IG0000014	creator	image	Food	183	6	10	3997	6107	0.04
IG0000015	brand	reel	Lifestyle	333	10	15	5240	6377	0.06
IG0000016	creator	reel	Music	271	8	13	5981	8127	0.04
IG0000017	creator	image	Travel	138	1	4	4489	5263	0.03
IG0000018	creator	carousel	Beauty	261	5	16	6267	9464	0.03
IG0000019	creator	image	Lifestyle	120	9	5	8187	11264	0.01
IG0000020	brand	carousel	Travel	39	0	2	3345	4755	0.01
IG0000021	creator	image	Music	303	0	12	3950	5435	0.07
Total				8629320	255649	432784	188167991	254000108	1,263.18

TABLEAU

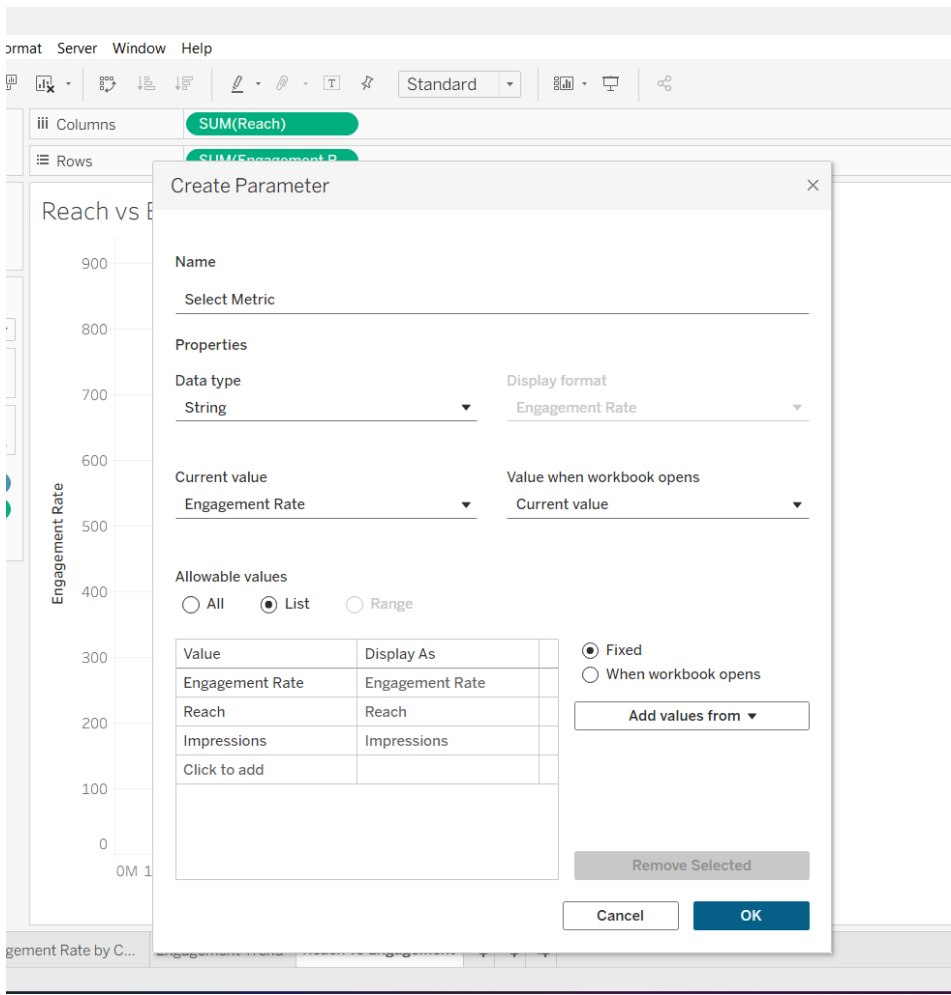
Import the data from sql server



FINAL DASHBOARD :



Build parameters



Selected Metric

×

```
CASE [Select Metric]
WHEN "Engagement Rate" THEN [engagement_rate]
WHEN "Reach" THEN [reach]
WHEN "Impressions" THEN [impressions]
END
```

The calculation is valid.

Apply

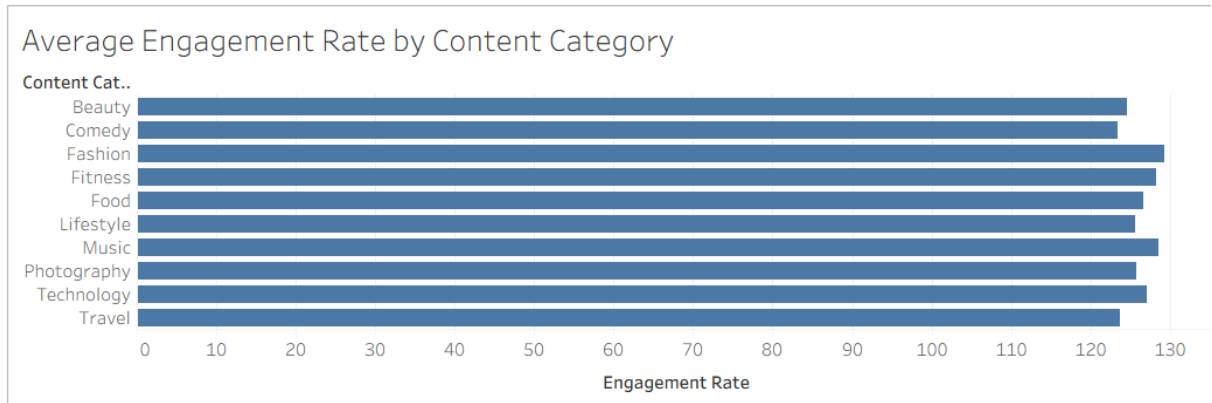
OK

Create a table to display post by account type

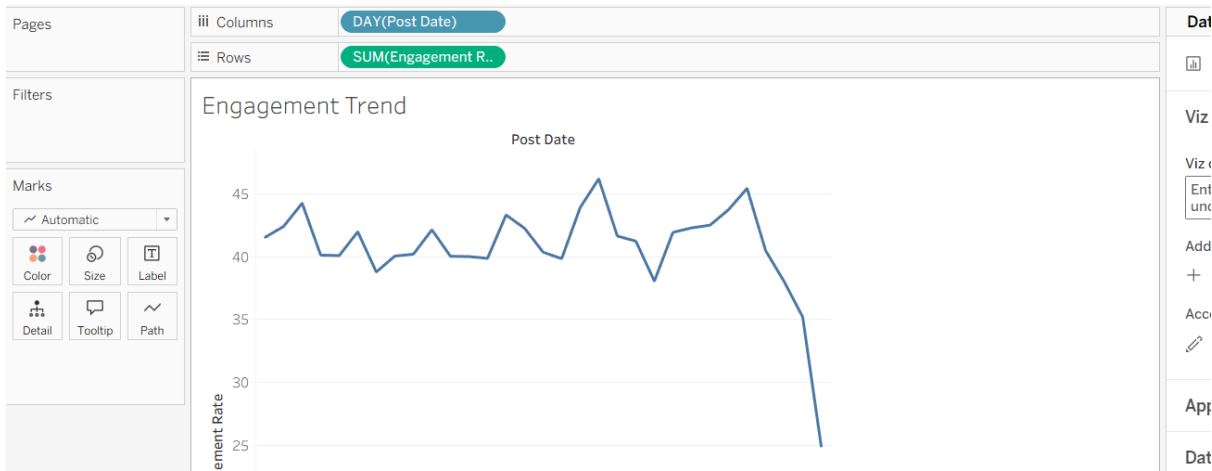
Posts by Account Type

Post Id	Account Type	
	brand	creator
IG0000001	Abc	
IG0000002		Abc
IG0000003	Abc	
IG0000004		Abc
IG0000005		Abc
IG0000006	Abc	
IG0000007		Abc
IG0000008	Abc	
IG0000009	Abc	
IG0000010		Abc
IG0000011		Abc
IG0000012		Abc
IG0000013	Abc	

to display average engagement rate by content category

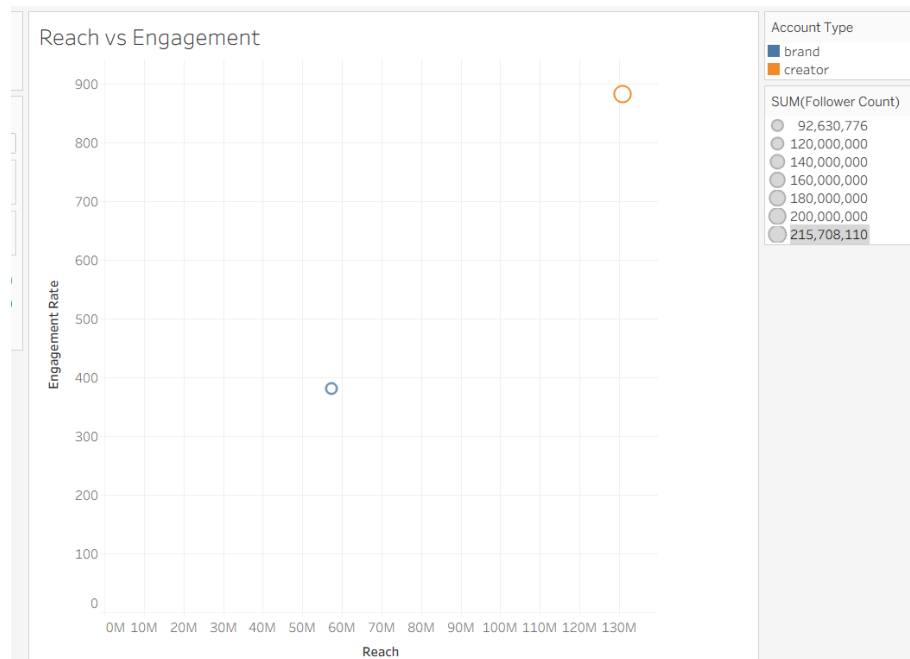


Line chart to visualise engagement trend with drill down



Year	2015
Quarter	Q2
Month	May
Day	8
More	▶

Reach VS Engagement Rate



Observed Insights

1. The dashboard contains approximately 30K Instagram posts with a total reach of 188M.
The average engagement rate is approximately 0.04.
This indicates a large volume of content and significant audience reach.
2. Creator accounts contribute a larger share of posts than brand accounts.
Creators account for approximately 69.82% of the posts, while brands account for about 30.18%.
This shows that creator accounts dominate the dataset.
3. Engagement varies across content categories, with Fashion showing the highest engagement.
Music and Fitness also show relatively high engagement compared with other categories.
Comedy and Travel appear comparatively lower in the displayed chart.
4. Engagement rate changes across the quarters, reaching its highest level in Q4.
The trend increases from Q1 to Q2, decreases during Q3, and rises again in Q4.
This indicates noticeable variation in engagement performance throughout the year.